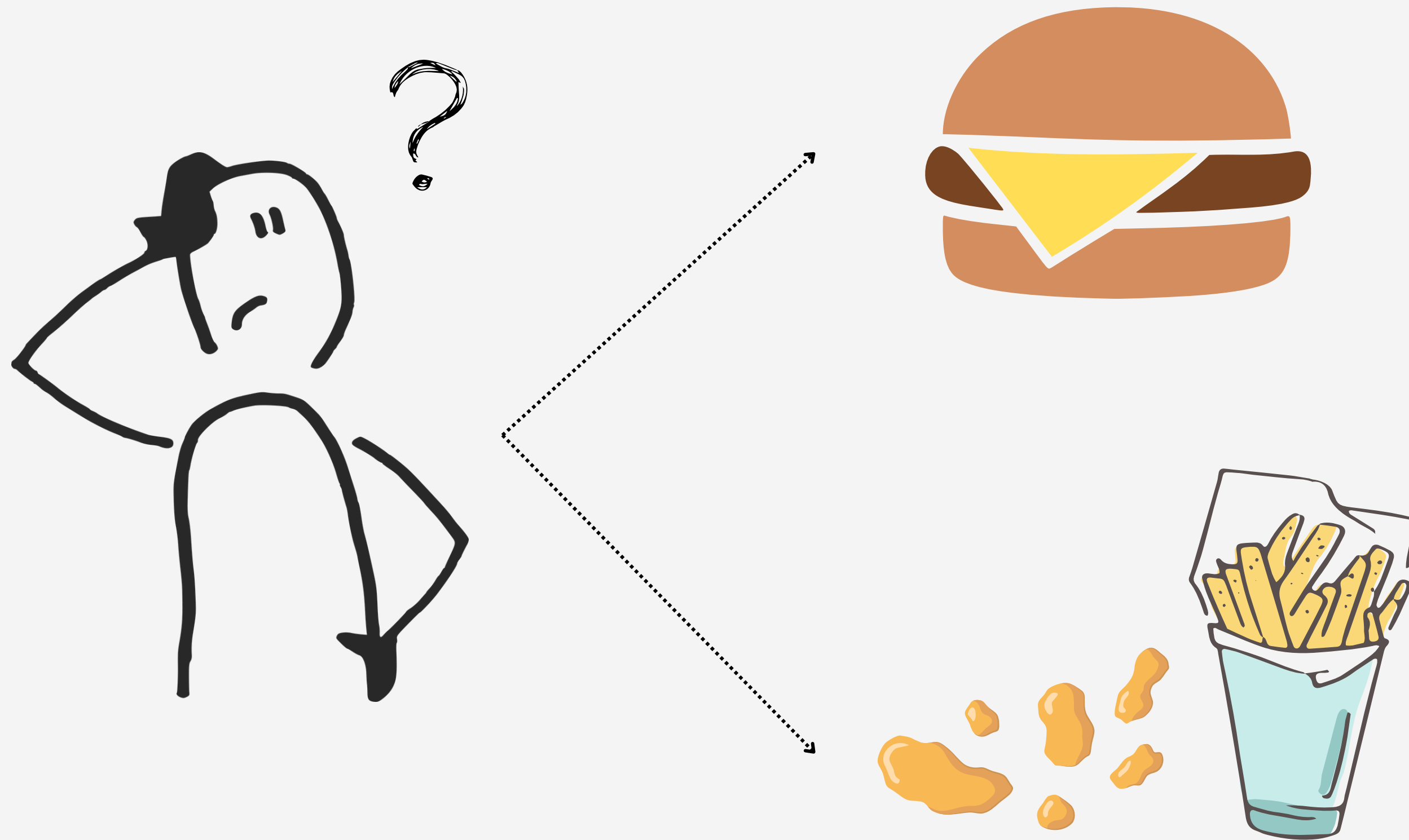


WHAT IF THAT C.ELEGANS COULD ACTUALLY THINK ?



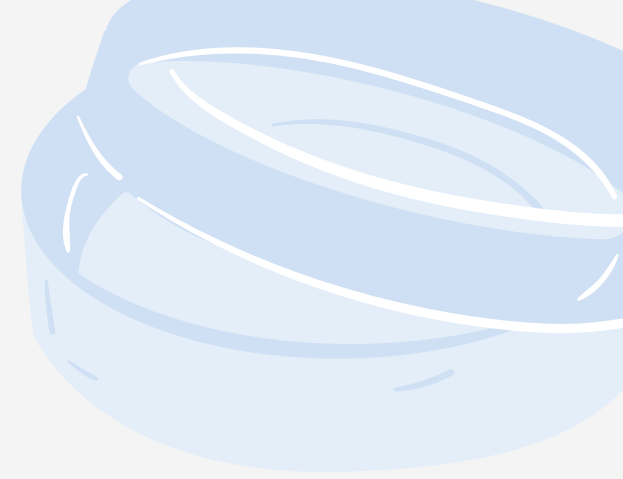


DO VERY SIMPLE ORGANISMS FOLLOW SIMILAR STRATEGIES WHEN THEY
MOVE BETWEEN FOOD PATCHES ?

BEHAVIORAL OPTIMIZATION: TESTING OPTIMAL FORAGING THEORY

Nour Saad, Oceane Alvarez Benita Ibrahim M1 BBS

SUPERVISORS : Alfonso Pérez Escudero and Roger Lloret Cabot



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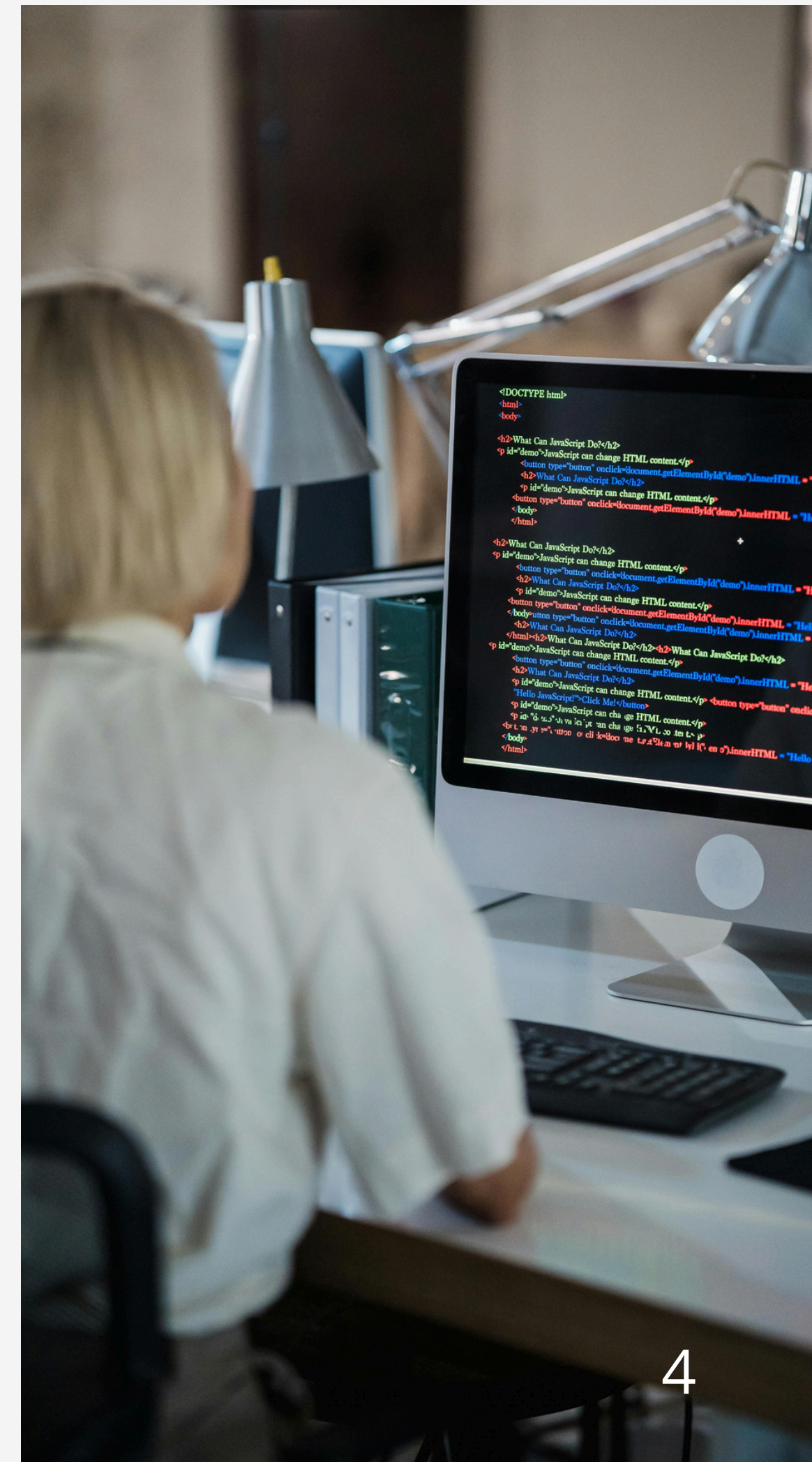
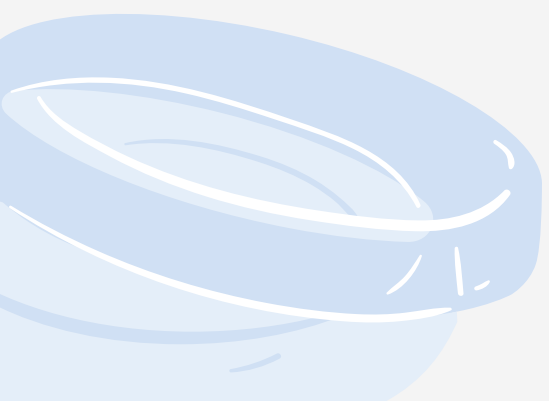
- Procedure
- Analysed the dataset

03 Results

- What we obtained ?

04 Discussion

- Interpretation



Introduction

- **The Problem:** Why do organisms leave a patch of food?
- **Significance:** Understanding the mathematics of efficiency in the context of *C.elegans* worms.



Methodology

Phase 01
Overview of the dataset



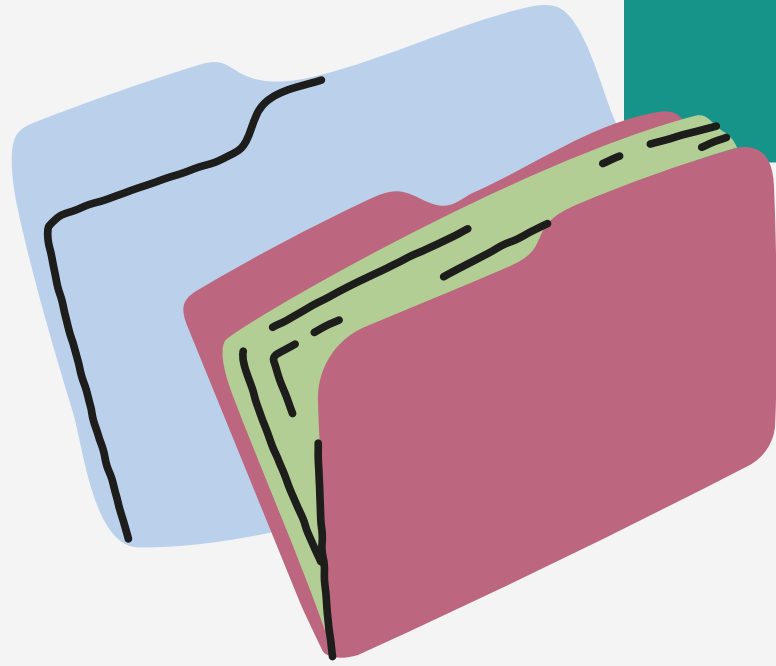
Phase 02
Creation of a Clean Dataset

Phase 03
Analysis of Multiple Worms

Phase 04
Speed Analysis

PHASE 01

Overview of the dataset



Minipatches_light_20260116 :

- containing experimental folders (that represents various videos)
- That contain many files



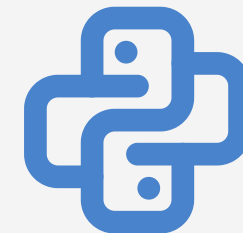
PHASE 02

Creation of a Clean Dataset



inconsistent if:

- traj.csv is missing but traj.mat is not empty.
- foodpatches_reviewed.mat is missing but traj.mat is not empty.



A Python script :

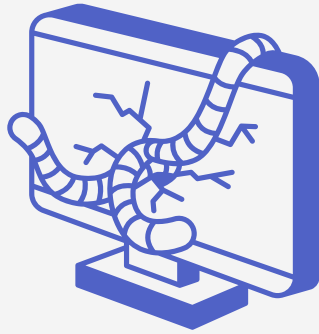
- Checks file existence.
- Measures file size -> empty files.
- Labels each video as:
OK
NO WORM
WEIRD



PHASE 03

Analysis of Multiple Worms

For each csv file:



Computation of
the proportion of
frames containing
1+ worms

Manual
inspection :
ambiguous
cases.



Assign labels:

- 0 worms
- 1 worm
- 1 worm with tracking errors
- 2+ worms



PHASE 04

Speed Analysis

For each csv file:

Computation of speed along trajectories

$$\text{Speed} = \sqrt{(xb - xa)^2 + (yb - ya)^2}$$

Plotting speed (y) over time (frames) (x)

To detect suspicious jumps => tracking errors/data corruption

Flag abnormal segments => exclusion/ correction

RESULTS